

Isolating Support Withdrawal in CALL: A Design Framework for Supported Two-Phase Reading (STR)

Abstract: Digital reading environments in computer-assisted language learning (CALL) widely integrate in-flow linguistic support (e.g., glosses, translations, annotations). While such support facilitates comprehension, its withdrawal is rarely implemented as a reproducible, system-level transition. In many platforms, assistance remains continuously available or is reduced informally, making transitions to unsupported reading difficult to standardize and inspect. This paper introduces Supported Two-Phase Reading (STR), a deployable task design that operationalizes categorical support withdrawal as a discrete, passage-level transition within a single reading sequence. STR structures reading into a supported phase followed by immediate unsupported rereading of the identical text after system-enforced cue removal. By localizing withdrawal at a deterministic boundary, STR transforms support availability into a controllable task-state variable within digital environments. A minimal web-based implementation with adult EFL learners ($N = 10$) demonstrates technical feasibility, enforceability of the transition, and protocol integrity. STR provides a reproducible design template for integrating scaffold withdrawal into LMS-based and web-based reading systems and for examining support transitions under controlled conditions.

1. Introduction

Digital reading systems in computer-assisted language learning (CALL) routinely incorporate in-flow linguistic support—glosses, dictionaries, annotations, and translations—to enable learners to engage with texts beyond their unaided proficiency. Although such support is easily embedded into platforms, its systematic withdrawal is not. In practice, support often remains continuously accessible, or its reduction depends on instructor judgment, learner choice, or adaptive algorithms that are difficult to standardize across contexts.

This creates both a practical and a design problem. When assistance is persistently available, transitions to unsupported reading are not instantiated as clearly defined task states. Unsupported processing becomes optional, delayed, or distributed across sessions, making it difficult to reproduce the same instructional configuration across classrooms, platforms, or studies.

Supported Two-Phase Reading (STR) addresses this implementation gap by formalizing scaffold withdrawal as a mandatory, binary transition embedded within a single reading sequence. Rather than treating support as a static interface feature, STR organizes it as a temporally bounded parameter under system-level control. Reading proceeds in two consecutive phases: a supported phase with full in-flow assistance, followed by immediate unsupported rereading of the identical passage after categorical removal of all designated cues.

While two-pass reading formats are pedagogically familiar, the transition between phases is typically left to instructor discretion or learner choice and is rarely implemented as a system-enforced, log-verifiable design constraint.

The contribution of STR is architectural and implementation-oriented. It specifies minimal structural requirements—passage-level scope, deterministic transition point, text invariance, and within-cycle irreversibility—under which withdrawal can be embedded reproducibly in digital reading environments. By defining a clear boundary between supported and unsupported states, STR enables platforms to enforce and log transitions without altering textual content or relying on cross-text progression.

In this sense, STR functions as a design template rather than a teaching method. It does not prescribe content, pedagogy, or learning theory alignment. Instead, it provides a reproducible configuration through which support availability can be parameterized and inspected within existing digital infrastructures.

1.1 Implementation Challenges in Higher-Level Extensive Reading

In higher-level Extensive Reading (ER), maintaining fluent comprehension typically requires approximately 95–98% lexical coverage. As texts approach the limits of independent reading capacity, learners increasingly rely on embedded glosses, pop-up dictionaries, or external lookup tools. In digital environments, such tools are often continuously accessible by default.

While persistent support stabilizes comprehension, it complicates structured transitions toward unsupported

reading. Instructors may encourage reduced reliance on assistance, yet most digital systems do not provide a simple mechanism for enforcing categorical withdrawal within a controlled sequence. As a result, unsupported reading remains optional or inconsistently implemented.

STR offers a practical mechanism for addressing this challenge. By presenting the same passage twice—first with full gloss visibility and then with all support categorically disabled—platforms can integrate a reproducible withdrawal cycle without modifying content or deploying complex adaptive algorithms. This bounded two-phase configuration enables systematic reintroduction of unsupported processing within the same discourse episode while preserving analytic transparency.

2. Support Withdrawal in CALL Reading: Design Background and Methodological Gap

Research in computer-assisted language learning (CALL) has extensively examined the role of linguistic support—such as lexical glosses, annotations, pop-up dictionaries, and integrated translations—in facilitating reading comprehension. Empirical findings generally indicate improved understanding while such support remains available. Despite this body of research, support is often treated as a persistent interface feature rather than as an explicitly specified design variable. In many studies, assistance remains continuously accessible or is contrasted across conditions without systematic attention to how the transition between supported and unsupported states is structurally defined. The methodological issue, therefore, concerns not the presence of support itself, but the manner in which its withdrawal is operationalized within a reading event.

Configurations of support withdrawal can be differentiated along at least three structural dimensions: (1) how clearly the withdrawal boundary is delimited, (2) whether textual material remains invariant across conditions, and (3) how temporally proximate supported and unsupported processing are. Designs that remove support across different texts or sessions (e.g., pretest–posttest comparisons or graded progressions) conflate withdrawal with text change and temporal delay. Item-level fading approaches selectively reduce gloss availability across encounters but do not define a unified passage-level transition within a single reading episode. Two-pass practitioner formats often incorporate supported reading followed by unsupported rereading; however, the transition between these states is not consistently specified as an enforceable, system-level constraint.

Across these configurations, support removal tends to be distributed, text-variant, or only partially localized. As a result, passage-level cue removal under text invariance is rarely operationalized as a clearly bounded architectural condition within a single reading sequence.

STR addresses three recurring design indeterminacies in CALL reading research:

- (1) temporal conflation, where withdrawal is separated from supported processing by time or text change;
- (2) control ambiguity, where support remains optional or adaptively variable; and
- (3) analytic fragmentation, where withdrawal operates at the item level rather than at a unified passage boundary.

By localizing a mandatory transition within a text-invariant episode, STR renders unsupported processing structurally determinate.

2.1 STR as a Minimal Withdrawal Specification

Supported Two-Phase Reading (STR) addresses this gap by formalizing withdrawal as a passage-level transition embedded within a single reading episode. The configuration rests on four structural constraints: passage-level scope, constant textual input across phases, a temporally localized shift, and within-cycle irreversibility.

In Phase I, all designated support remains available throughout the passage. At a predefined point, the system removes support categorically, and Phase II consists of immediate rereading of the same text under altered access conditions. Because the shift is singular and system-enforced, it operates as a discrete design event rather than as a gradual or distributed adjustment.

While scaffold withdrawal is pedagogically familiar, its implementation often varies in boundary definition and textual continuity. STR establishes an explicit configuration in which cue removal occurs without altering discourse input, allowing transition-related measures to be attributed to support availability rather than to text change or temporal separation.

By treating withdrawal as a controllable task parameter within a unified episode, STR enables systematic

comparison of different fading regimes (e.g., binary, continuous, adaptive) under constant input conditions. The framework therefore functions as research infrastructure rather than as a prescriptive instructional model.

Design Property	Continuous Gloss	Distributed Item-Level Removal	Gradual/Adaptive Fading	STR
Withdrawal scope	None (support persistent)	Item-level	Mixed (item/passage)	Passage-level
Temporal localization	Not applicable	Distributed	Progressive	Single discrete event
Deterministic transition point	No	No	Typically no	Yes
Within-cycle irreversibility	Not applicable	No	No	Yes
Text invariance across withdrawal	Not required	Not required	Not required	Required
Passage-level boundary identifiable	No	No	Indeterminate	Yes
Observable phase differentiation	No discrete boundary	No discrete boundary	Non-atomic transition	Yes

Table 1. Structural comparison of withdrawal configurations

As shown in Table 1, STR differs not in the inclusion of support, but in the structural organization of its withdrawal. It specifies a deterministic, passage-level boundary within a text-invariant reading episode, providing a reproducible architectural reference configuration.

3. Conceptual Positioning of Supported Two-Phase Reading

Supported Two-Phase Reading (STR) is proposed as a task-level design specification for digital reading in CALL. Rather than defining a teaching method or a theory of learning, STR specifies how support withdrawal is structurally instantiated within a single reading episode.

Within this framework, unsupported processing is treated as a temporarily enforced task condition rather than as a learner trait or delayed outcome. The analytic focus shifts from whether learners are autonomous readers to how reading unfolds when unsupported processing becomes the only available task state. Autonomy is therefore operationalized as a design-defined requirement within the task, not as an inferred learner characteristic.

3.1 STR as a Design-Level Specification

STR specifies a fixed two-phase structure applied to identical textual material. In Phase I (Supported State), the text is presented with full in-flow support. In Phase II (Unsupported State), the identical text is presented with all designated support categorically removed.

The transition between phases is mandatory, binary, and unidirectional. Once Phase II begins, supported input is structurally unavailable within the same cycle. This shift is enforced at the system level rather than negotiated through instruction, learner choice, or adaptive algorithms. Unsupported processing is therefore instantiated as a required task condition within the same episode.

Phase II reflects task engagement under altered support constraints. The framework treats this immediate unsupported processing as a design-defined state, without implying internalization, retention, or long-term autonomy.

3.2 Differentiation from continuous-support formats

STR occupies a clearly bounded position on the scaffold-fading continuum and differs from formats in

which assistance remains continuously accessible. Many digital and print environments provide integrated annotations or learner-controlled gloss access throughout reading, leaving withdrawal optional, distributed, or deferred.

The distinction is structural. STR defines a temporally bounded support phase followed by categorical, irreversible removal within the same episode. When support remains continuously available, unsupported processing is not instantiated as a required task state within a text-invariant sequence.

Because Phase II involves immediate rereading of identical textual material, observed differences may reflect both familiarity effects and categorical withdrawal. Research on rereading suggests that second encounters often shift processing toward discourse-level representations (Grabe, 2009; Raney, 2003). STR does not attempt to separate rereading from withdrawal; rather, it specifies their joint occurrence under a deterministic boundary. Exposure-matched continuous-support rereading conditions (Supported → Supported) may serve to differentiate familiarity effects from withdrawal-related contrasts.

System-level enforcement of task conditions has been increasingly examined in CALL research through the analysis of interaction logs. Hui et al. (2023) demonstrate how system-generated logs can serve as verifiable indicators of task execution and learning processes. In this sense, STR's categorical withdrawal boundary constitutes a log-detectable task state transition, enabling process-level inspection of support availability within digital environments.

3.3 Why Binary Rather Than Gradual Fading

The binary transition in STR is chosen for methodological clarity rather than pedagogical superiority. Gradual or adaptive fading distributes support reduction across time, items, or performance thresholds, diffusing the boundary between assisted and unassisted states.

By contrast, STR establishes a discrete and reproducible withdrawal point within a single reading episode. The shift from full support to zero support occurs at a clearly defined boundary, creating a determinate state change. This configuration enhances comparability and analytic tractability, allowing STR to function as a minimal baseline against which gradual or adaptive fading models may be evaluated.

3.4 Minimal State Specification

STR defines a two-state task structure: Phase I presents the text with complete in-flow support, and Phase II presents the identical text with all designated support removed. This structure rests on four constraints: text invariance across phases, full support in Phase I, categorical removal in Phase II, and irreversibility of the transition within a single cycle. Configurations involving partial support, learner-controlled reversal, or gradual within-cycle fading fall outside this specification.

Within this design, the Phase I → Phase II shift constitutes a discrete system-level state change. Withdrawal is thus instantiated as a bounded task variable, without implying cognitive isolation or guaranteed learning gains.

Each STR cycle is architecturally self-contained: one supported reading, one categorical withdrawal event, and one unsupported rereading of identical text. Cumulative or longitudinal effects, while potentially researchable, remain external to the architectural definition. This closure preserves analytic clarity at the cycle level.

4. STR Architecture

4.1 Core Structural Components

Figure 1 depicts a single STR sequence consisting of two task states—supported reading (Phase I) followed by unsupported rereading (Phase II)—connected by a mandatory, one-way removal of linguistic support. Once Phase II begins, support cannot be reinstated within the same sequence.

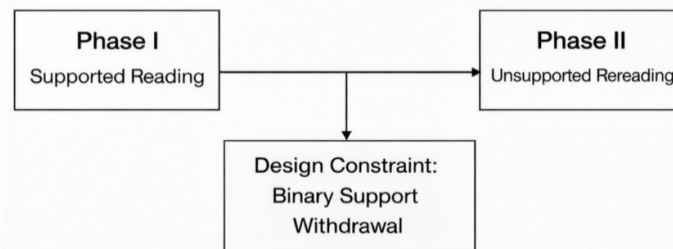


Figure 1. Binary support withdrawal in Supported Two-Phase Reading (STR).

In typical classroom implementations (e.g., A1–B1 learners), STR is applied to short, coherent passages (approximately 150–200 words), allowing both phases to occur within one session while preserving discourse continuity. Multiple cycles may be sequenced across larger text sets; however, each cycle remains structurally independent and contains a single state transition.

Phase I: Supported Reading. Learners read a passage with complete in-flow support (e.g., lexical translations or grammatical annotations) embedded directly in the text. All designated support remains accessible throughout this phase.

Transition. At a predefined boundary, the system withdraws all support simultaneously. The shift from full access to complete absence is categorical and irreversible within the sequence. Designs involving partial gloss retention, adaptive fading, or gradual reduction are not included in this specification. The transition marks a clearly defined change in task conditions.

Phase II: Unsupported Rereading. Learners immediately reread the identical passage without access to support. Because the discourse input remains constant, the sole structural difference between phases is support availability, eliminating confounds related to text substitution or temporal separation.

This two-phase configuration defines the operational core of STR. The withdrawal boundary is system-enforced and log-verifiable, enabling consistent implementation across platforms without reliance on instructor judgment or learner control. As such, STR provides a reproducible design configuration for examining categorical support removal within digitally mediated reading environments.

4.2 Operational Definition of Support

Within STR, support refers to any linguistic or metalinguistic information integrated directly into the primary text stream that meets the following criteria:

- Externally provided (not generated by the learner),
 - Immediately accessible without navigation or lookup actions,
 - Intended to reduce processing demands or facilitate immediate text access.
- Common instantiations include lexical glosses (e.g., L1 translations), grammatical annotations, brief explanatory notes, or phonological guidance.

STR does not prescribe which types of support must be included. The specific composition of support remains an implementation parameter. What the architecture requires is categorical symmetry: any support layer available in Phase I must be entirely absent in Phase II within the same cycle.

This symmetry ensures that the only structural difference between phases is the presence or absence of in-flow support.

5. Methodological Validation (Protocol Instantiation)

This section reports a methodological validation of the Supported Two-Phase Reading (STR) architecture through instantiation in a minimal web-based reading environment. The purpose of this validation is architectural rather than instructional.

The validation addresses three design-level questions:

Can the STR architecture be reliably implemented in a digital system?

Is the mandatory Phase I → Phase II transition consistently enforced?
 Do participants complete STR cycles without protocol violations or system-level circumvention?
 The validation does not evaluate learning outcomes, retention, comprehension gains, or transfer. Those questions require separate empirical designs and lie outside the scope of this architecture-focused contribution.

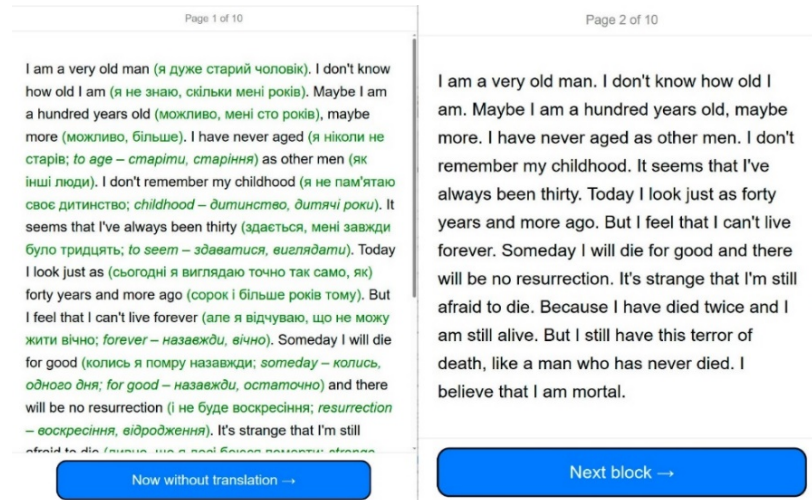


Figure 2. Interface states in the STR protocol instantiation.

The left panel (Phase I) displays supported reading with in-flow assistance integrated directly into the text.
 The right panel (Phase II) displays immediate unsupported rereading of the identical text following system-imposed withdrawal of all support.
 The transition is unidirectional and cannot be reversed within a single STR cycle.
 All participants completed all 5 STR cycles without technical failure or task abandonment. The system architecture did not permit bypassing or reversing the Phase I → Phase II transition. Interaction logs confirmed correct sequencing of phases and consistent enforcement of protocol constraints across sessions.

A public demonstration of the instantiated STR protocol is available for inspection:

<https://str-study.github.io/supported-two-phase-reading/>

5.1 Protocol-Bound Descriptive Indicators

In line with the instrument-validation focus, only minimal indicators necessary to confirm architectural execution were recorded. These included:

- Completion of all STR cycles (protocol adherence),
- Non-zero dwell time on Phase II pages (confirmation of unsupported engagement),
- Reading time per phase, automatically recorded in seconds.

Across 50 completed STR cycles, mean reading time in Phase I was 122.6 s (SD = 77.9), and mean reading time in Phase II was 49.6 s (SD = 26.2).

These values are reported descriptively to document phase differentiation within the implemented architecture. No inferential analysis is performed, and no interpretation is made regarding proficiency, fluency, or learning effects. Differences in reading time may reflect multiple factors, including absence of visible support, rereading of familiar content, task pacing, or fatigue.

The purpose of reporting these measures is limited to confirming that the two phases were operationally distinct and sequentially executed within the system.

5.2 Scope and Limitations of the Present Validation

The present validation establishes implementability and protocol integrity at the design level. It does not

assess depth of engagement, comprehension accuracy, or cognitive processing during Phase II.

Verification of qualitative differences in processing would require additional methodologies, such as comprehension assessment, eye-tracking, or concurrent verbal protocols. Such methods fall beyond the scope of architectural validation and are deferred to future empirical investigation (see Section 8).

A brief post-task check confirmed detectability of the Phase I → Phase II transition as a change in task conditions. These responses served solely to verify perceptibility of the withdrawal boundary and were not subjected to thematic or inferential analysis.

6. Analytic Value of STR for CALL and Educational Media Research

The contribution of Supported Two-Phase Reading (STR) is methodological. Rather than introducing support withdrawal as a new pedagogical practice, STR specifies cue removal as a discrete, passage-level event within a single, text-invariant reading episode. In many fading and glossing designs, support reduction is distributed across items, sessions, or texts. Under such configurations, withdrawal is entangled with text change, temporal delay, or partial support persistence, making it difficult to isolate cue removal as an analytically bounded variable.

STR defines a configuration in which cue removal occurs at a single deterministic boundary while the discourse material remains constant. Supported and unsupported processing therefore unfold over identical textual input within the same session. This renders cue availability—not text difficulty, exposure variation, or distributed fading—the primary structural contrast between phases.

STR does not eliminate all sources of variation—familiarity and rereading effects remain present—but it removes the structural confounds introduced by cross-text comparison or progressive support reduction. Withdrawal becomes a clearly localized state transition rather than a gradual interface property.

For CALL research, this configuration enables within-episode contrasts under controlled structural constraints, allowing cue removal to be examined independently of text change. For educational media research, STR formalizes support availability as a boundary-defined interface parameter, permitting systematic comparison with gradual or adaptive fading models.

Because the withdrawal boundary is discrete and passage-level, STR enables boundary-anchored observation of phase-level contrasts—such as latency shifts, interaction reorganization, or differentiated engagement patterns—that are difficult to specify in configurations without a clearly delimited state transition. In this sense, STR contributes analytic inspectability rather than instructional claims: it renders cue removal a structurally identifiable event within digital reading environments. As a bounded, non-adaptive configuration, STR may also function as a stable reference condition for comparative examination of more complex fading architectures.

7. Instructional Problem and Scalable Implementation

Digital reading environments typically maintain continuous in-flow support to stabilize comprehension. In classroom practice, transitions from assisted to independent reading are often implemented informally and may vary across instructors, groups, or sessions. As a result, unsupported processing is not always instantiated as a clearly defined task state within otherwise comparable reading activities. Scaffold fading may occur gradually, optionally, or through teacher-managed decisions, leaving the moment of full withdrawal structurally diffuse.

STR addresses this design-level constraint by embedding withdrawal directly into task architecture. The Phase I → Phase II transition reorganizes reading as a bounded micro-sequence in which unsupported processing is categorically instantiated within the same text. Rather than depending solely on instructional discretion, the architecture guarantees that each cycle contains a deterministic unsupported phase. In classroom integration, this establishes a repeatable instructional unit—supported access followed by system-enforced withdrawal—while leaving pedagogical sequencing and text selection under teacher control.

From a scalability perspective, STR separates support generation from withdrawal enforcement. In static publishing formats (print or e-book), Phase I may include integrated annotations, while Phase II presents the identical text without support, preserving a fixed and visible boundary. In adaptive digital environments, support density or gloss type in Phase I may vary across learners, whereas categorical withdrawal in Phase II remains invariant as a structural condition.

By defining a minimal, reproducible passage-level fading unit, STR enables controlled cross-classroom and

cross-platform comparison of withdrawal configurations without presupposing instructional superiority. The specification does not assert improved learning outcomes; rather, it establishes a consistent architectural condition under which supported and unsupported task states can be systematically examined relative to continuous-support formats.

8. Scope and Future Directions

This paper presents STR as a task-level specification for categorical support withdrawal in digital L2 reading. The framework defines how unsupported processing can be instantiated as a bounded state within a single reading sequence, but it does not evaluate instructional effectiveness, retention, or long-term development. Phase II performance is treated as task-bound behavior under altered support conditions.

Future research may examine whether localized cue removal is associated with measurable differences in phase-level processing or delayed retention under exposure-matched conditions. Questions of transfer, durability, and autonomous reading development require separate empirical investigation beyond the present architectural validation.

9. Research Implications and Testable Extensions of STR

The present contribution is architectural in scope. STR specifies a categorical withdrawal boundary as a task-level state transition within a single, text-invariant reading episode. The framework itself does not entail specific learning outcomes; however, it establishes clearly defined task conditions under which theoretical interpretations may be examined empirically.

9.1 Experimental Preconditions

Interpretation of empirical findings presupposes several structural controls:

- (1) identical textual materials across comparison conditions,
- (2) exposure- and time-matched designs,
- (3) verified categorical removal of support in Phase II, and
- (4) alignment between text difficulty and learner proficiency.

Failure to meet these conditions constrains interpretability of results, as observed differences may otherwise reflect text variation, unequal exposure, or uncontrolled fading rather than withdrawal effects per se.

9.2 Conditional Hypothesis: Delayed Retention Under Categorical Withdrawal

From a retrieval-oriented perspective, Phase II requires learners to process previously supported lexical items without external cues, which may increase reliance on internally available representations compared to a continuous-support condition. Under exposure- and time-matched conditions, learners completing an STR sequence (Supported → Unsupported) may therefore exhibit different delayed productive recall profiles than learners in continuous-support configurations (Supported → Supported). Any such differences, if observed, would reflect variation in recall dynamics under structurally controlled conditions rather than assumed performance gains. Delayed generative measures—such as cued recall or productive gap-fill tasks—may serve as indicators for examining these potential profile distinctions.

If no systematic differences are observed under controlled conditions, retrieval-based interpretations of STR would require refinement. Such findings would inform theoretical accounts of Phase II processing without altering the architectural definition of categorical withdrawal.

9.3 Withdrawal Boundary as a Testable Design Event

Independent of retention outcomes, STR defines a discrete Phase I → Phase II transition within a single reading sequence, marking a formally specified change in task constraints: full in-flow support followed by categorical withdrawal.

If the transition functions as more than a nominal design feature, it should be empirically inspectable as a state shift. Under controlled conditions, such a shift may be reflected in phase-level contrasts, including changes in reading latency, interaction patterns, or perceived task difficulty.

Failure to detect systematic differentiation at the boundary would not invalidate the architectural specification, but would limit claims regarding its functional distinctiveness within observable task performance.

9.4 Structural Detectability as Initial Research Stage

A first empirical step in a broader STR research program concerns whether the Phase I → Phase II transition yields measurable differentiation within a single reading sequence. The guiding research question is: Does STR produce observable phase-level differentiation between supported and unsupported reading under constant discourse input?

This stage does not assess learning outcomes or delayed retention. Rather, it examines whether categorical withdrawal constitutes a functionally distinct task state. Under controlled conditions, relevant indicators may include shifts in reading latency, perceived task difficulty, or self-reported autonomy.

The objective is to determine whether the withdrawal boundary generates an experientially and behaviorally distinct shift in task constraints rather than remaining a purely formal design feature. A short-term implementation (e.g., 1–2 weeks) may be sufficient to establish whether consistent phase differentiation emerges across repeated STR sequences.

9.5 Architectural Testability and Scope

At minimum, STR advances a testable architectural claim: that categorical support withdrawal can be instantiated as a discrete, enforceable task state within a single text-invariant episode. The claim concerns preservation of a coherent state transition—full support in one phase and full absence in the next—rather than assumptions about cognitive mechanisms or guaranteed learning gains.

The broader instructional relevance of STR ultimately depends on whether empirical investigations conducted under these controlled structural conditions support particular theoretical interpretations of Phase II processing.

10. Conclusion

This paper has presented Supported Two-Phase Reading (STR) as a reproducible task architecture for implementing categorical support withdrawal in digital reading environments. By preserving text invariance while imposing a deterministic boundary, STR transforms cue availability from a persistent interface feature into a manipulable, system-enforced task-state variable.

Its contribution is methodological and implementation-oriented: the defined boundary enables within-episode comparison of supported and unsupported processing under exposure-matched conditions, without reliance on cross-text substitution or distributed fading. Withdrawal becomes not only observable but technically controllable within a defined reading sequence.

In practical terms, STR can be deployed in any LMS or web-based reader by presenting the same passage twice, with gloss visibility enabled in Phase I and categorically disabled in Phase II through system-level control. No content modification or adaptive sequencing is required; only deterministic toggling of support availability. This minimal requirement lowers the implementation threshold for integrating reproducible scaffold withdrawal into existing digital infrastructures.

STR does not compete with gradual or adaptive fading models; rather, it establishes a reference configuration against which such models can be compared. By fixing withdrawal as a deterministic, passage-level state variable, STR provides a stable baseline for implementing and analyzing scaffold removal in digitally mediated reading.

While no claims are advanced regarding instructional superiority or guaranteed learning gains, STR specifies the structural preconditions under which such claims can be meaningfully tested. As a passage-level and platform-independent configuration, it offers a controlled and deployable framework for integrating support

withdrawal into contemporary educational technologies.

Ethical Statement

A minimal-risk protocol instantiation involving 10 adult EFL learners was conducted to verify architectural implementability. No personally identifiable information was collected, and informed consent was obtained from all participants. In accordance with applicable local regulations, formal institutional ethical review was not required.

The author discloses prior commercial distribution (2017–2025) of reading materials employing a two-phase supported reading format. These materials were not used as part of the reported study. The present manuscript contains no promotional content and advances no claims regarding instructional effectiveness.

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